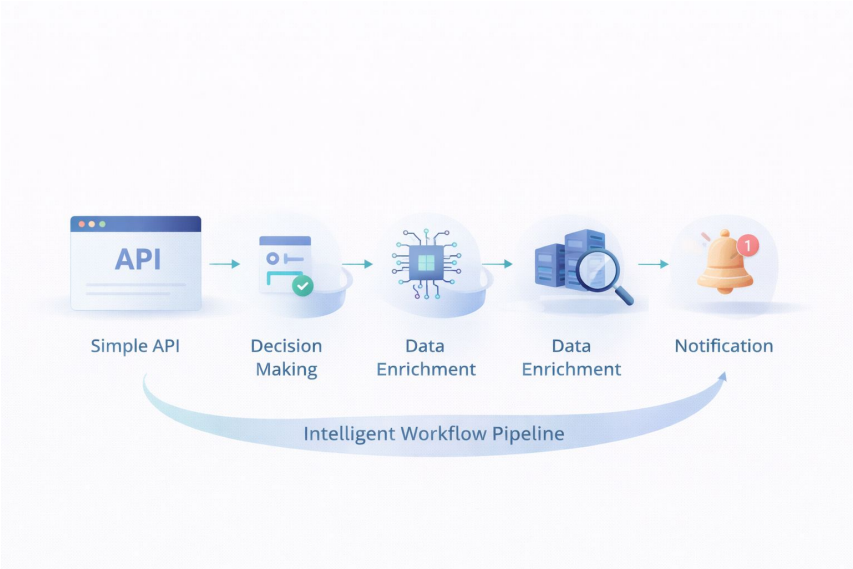
Python Programming Crash

Mini Agentic AI Framework

From API to Intelligent System (Agent Workflow)

Transform your API into a system that processes, decides, and responds.

- Move from simple API → smart workflow
- Chain multiple processing steps
- Prepare for agent-based systems
- Build incrementally before full framework
- Focus on system thinking



The Big Idea — Intelligent System = Pipeline

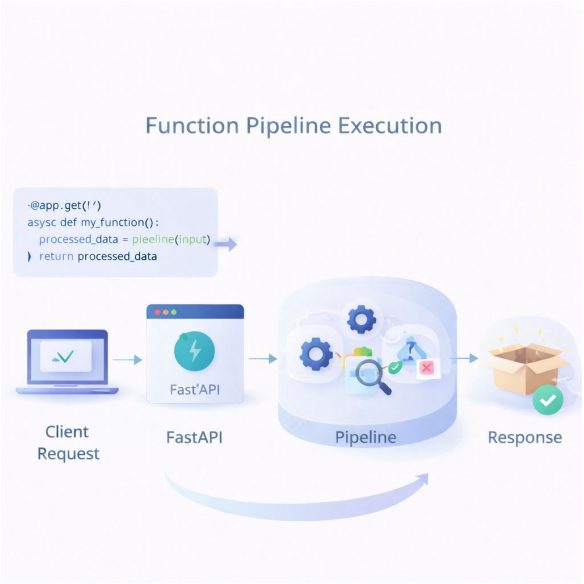
An intelligent system is just a sequence of simple steps.

- Input → Process → Decide → Output
- No magic involved
- Each step has a single responsibility
- Steps are chained together
- Logic flows sequentially

Our Workflow (Real Example)

Combine everything from previous days into a real workflow.

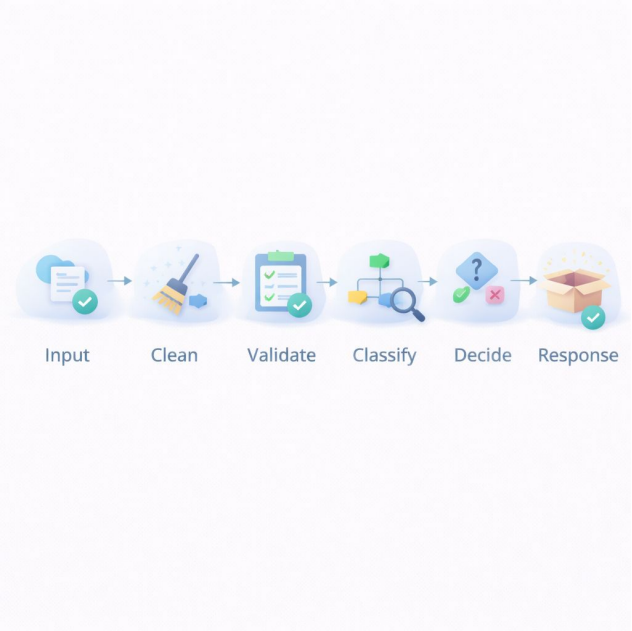
- User input enters system
- Unicode cleaning (Day 1)
- Validation (Day 2)
- Classification + decision (Day 3)
- Response returned



Step 1 — Build Simple Pipeline

Start with plain Python to understand the workflow.

- Clean input text
- Simulate AI classification
- Generate decision
- Return structured output
- Focus on flow, not architecture



Example code

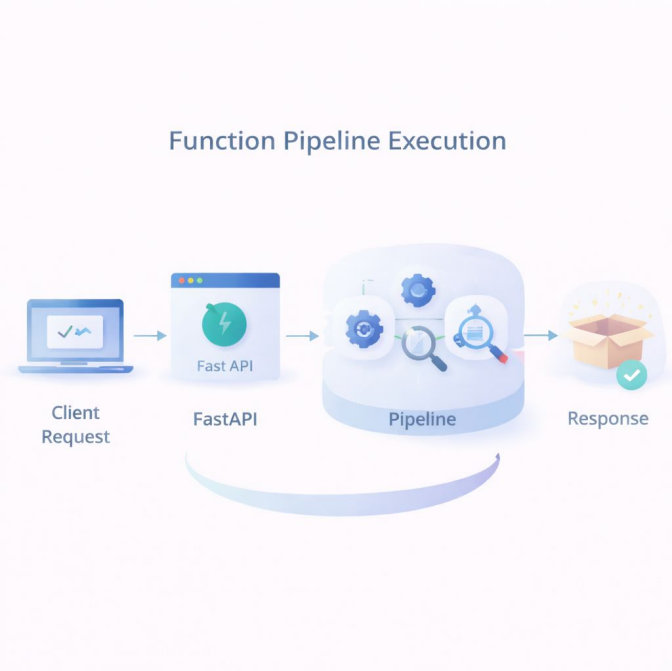
```
def simple_pipeline(text):
    cleaned = clean_unicode_text(text)
    category = mock_ai_call("classify", cleaned)
    decision = mock_ai_call("decision", category)
    return {
        "cleaned": cleaned,
        "category": category,
        "decision": decision
    }
```

→ No metaclass yet
→ No decorators yet

Step 2 — Expose Pipeline via API

Turn your pipeline into a usable API endpoint.

- Connect logic to FastAPI
- Accept user input
- Return structured response
- Reuse previous API knowledge
- Keep implementation simple



Exmaple code:

```
 @app.post("/simple-process")
def process(data: UserInput):
    return simple_pipeline(data.text)
```

→ Same API pattern as before

Step 3 — Test the Workflow

Run the full pipeline end-to-end.

- Input flows through system
- Each step transforms data
- Final decision is generated
- Output is structured
- Easy to test in Swagger

Example input

```
{
  "text": "urgent system error 🚨"
}
```

→ Feels like AI already

What Just Happened?

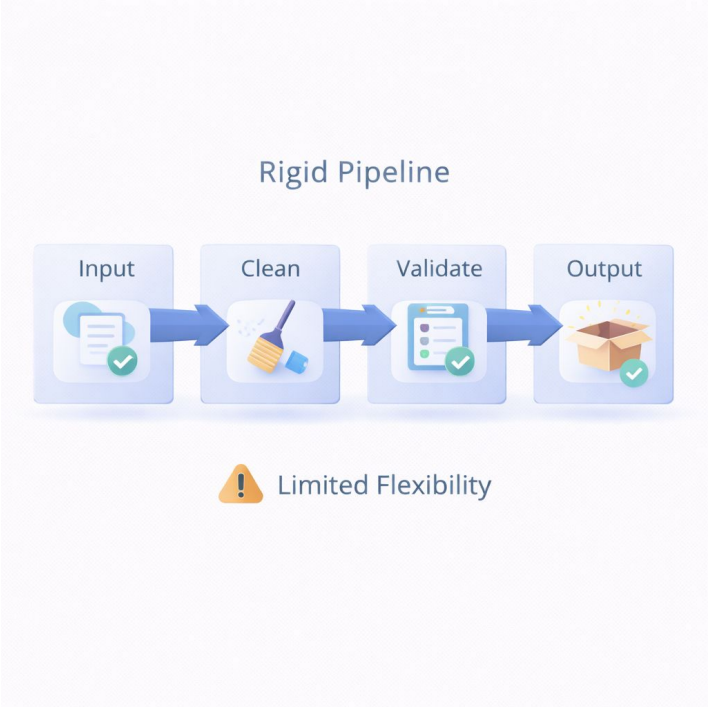
You built a simple intelligent system.

- Data cleaned and normalized
- Logic applied step-by-step
- Decision generated
- Response returned
- Full workflow executed

Limitations of Simple Pipeline

This approach works but doesn’t scale.

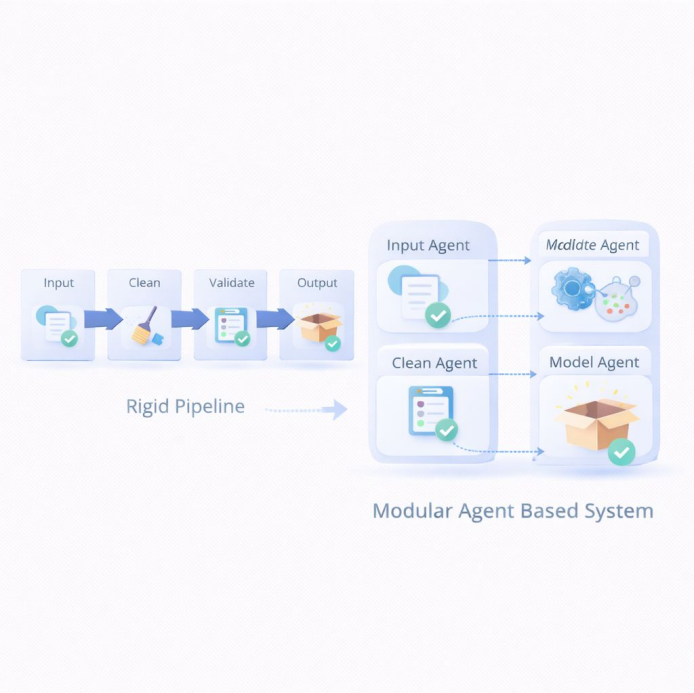
- Hardcoded pipeline flow
- Difficult to extend
- No modular components
- No reuse across systems
- Not suitable for large systems



What We Improve Next

Upgrade from simple pipeline to structured system.

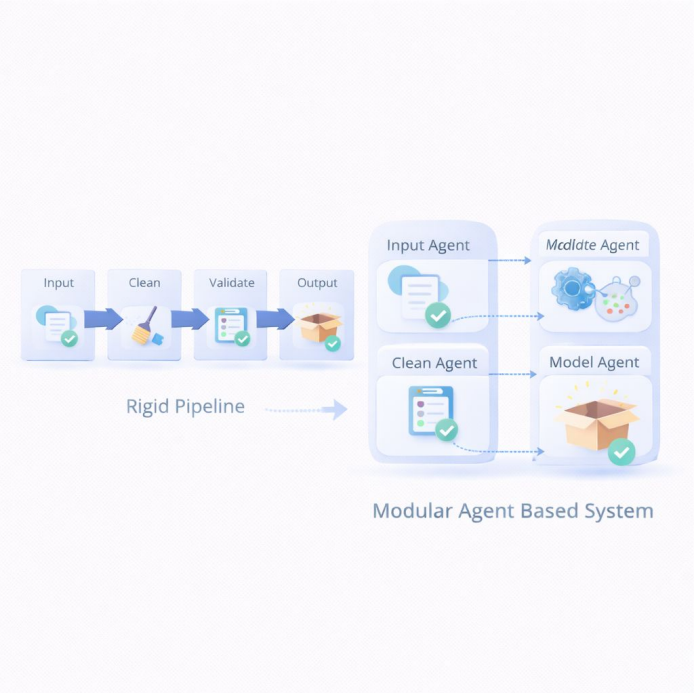
- Replace functions → agents
- Add decorators for safety
- Auto-register agents (metaclass)
- Introduce orchestrator
- Improve modularity



Simple vs Advanced Architecture

Same idea, different structure.

- Simple: function → function → function
- Advanced: agent → agent → orchestrator
- Logic remains the same
- Architecture becomes scalable
- Easier to extend and maintain



Connecting to Main Lab

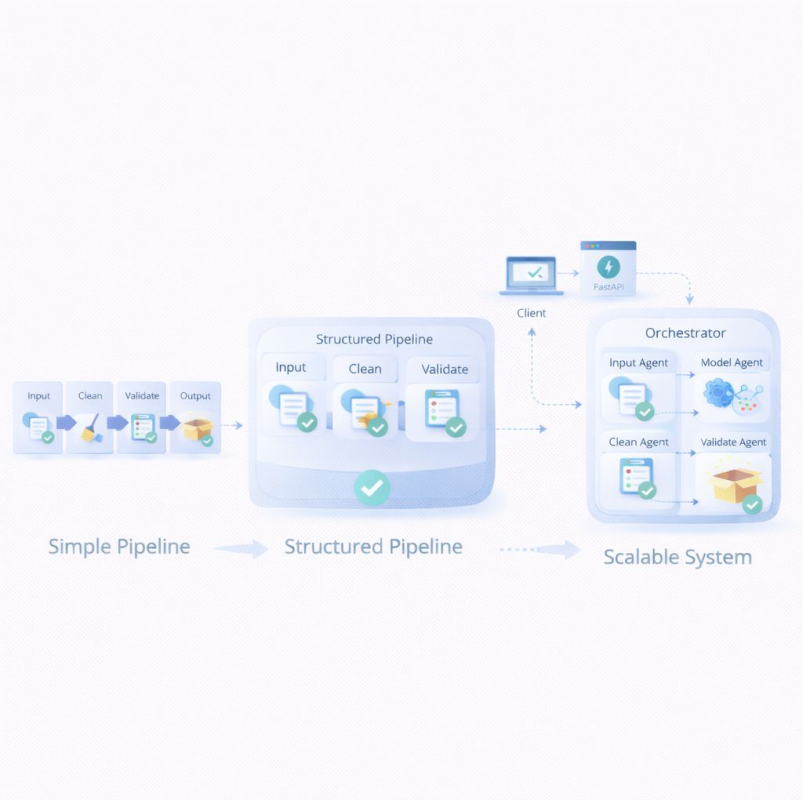
Prepare to build the full agent system.

- Agent-based architecture
- Dynamic workflow execution
- Logging and retry mechanisms
- API integration
- Real-world system design

Final Mindset

Build simple first, then scale.

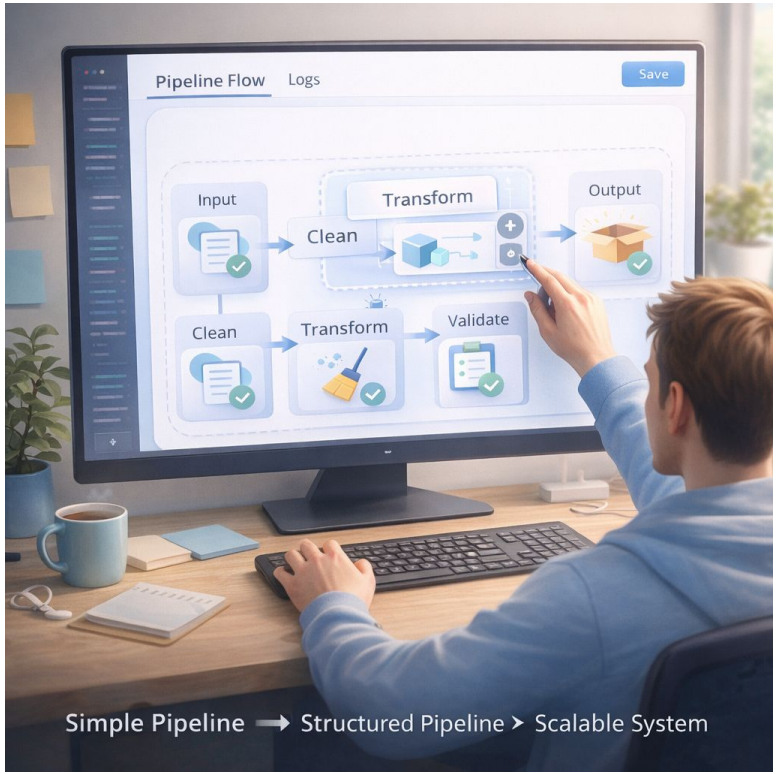
- Start with basic pipeline
- Understand the flow
- Add structure gradually
- Avoid overengineering early
- Think like a system designer



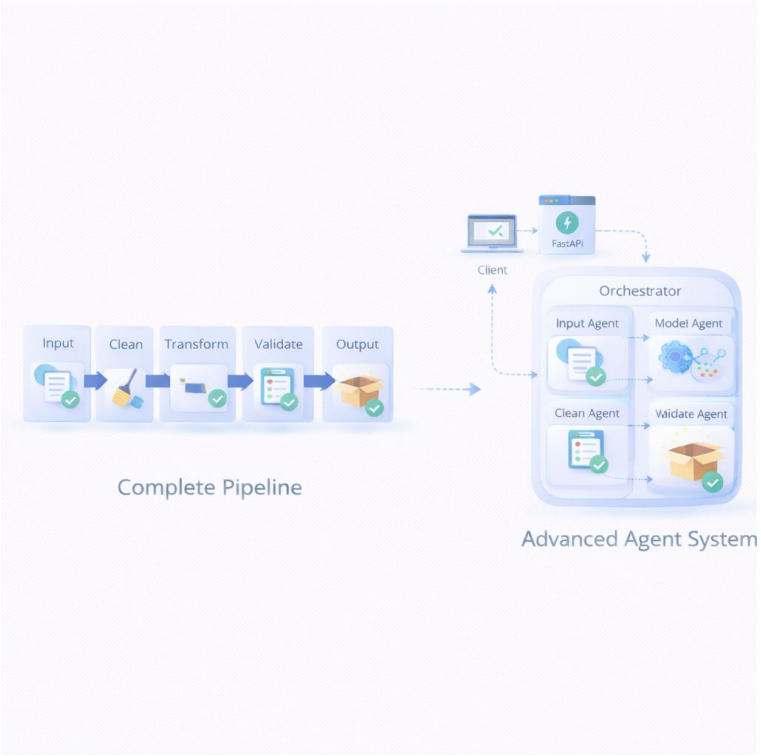
Mini Task — Try It Yourself

Experiment with your pipeline.

- Modify classification rules
- Change decision logic
- Add new processing step
- Test using Swagger
- Observe system behavior



Summary

You are ready to build agent systems.

- Built working pipeline
- Exposed it via API
- Simulated AI workflow
- Understood system flow
- Ready for full framework